GENESIS OS / RELATIONSHIP SCORES

E_GRIEF Official Companion Contract v1

Detailed engineering contract for Spiritual Depletion Index tiers, influence caps, packet schemas, runtime exposure rules, and rollout boundaries

Purpose	Merge the upgraded E_GRIEF module into the official build set as a governed companion system that can be implemented without guesswork.
System stance	E_GRIEF may adapt pacing, copy, feature exposure, and Sanctuary posture, but it may not freely mutate workbook score law.
Primary consumers	Sanctuary runtime, ExsAnonymous, Last Light, AI orchestration, readiness/forecast context, moderation-safe display surfaces.
Authority order	Workbook score law -> corrected master blueprint -> this companion contract -> module copy libraries and rollout configs.

This contract calibrates the grief system to the current Genesis OS architecture so the dev team can build it as a controlled runtime layer instead of a poetic side quest with hidden math.

1. Purpose, scope, and non-negotiable system stance

This specification promotes E_GRIEF into the official companion-system build set. It takes the upgraded Spiritual Depletion Index model and converts it into a production contract covering source inputs, packet schemas, SDI tiers, influence caps, runtime exposure rules, downstream write-backs, rollout boundaries, and safety posture.

- E_GRIEF is a companion-state engine for grief load, not a replacement for the core assessment, archetype, shadow, polarity, or pair-score engines.
- Its production purpose is adaptive care: slowing surfaces down, changing AI tone, shaping Sanctuary entry posture, reducing social pressure, and supplying governed readiness / forecast context when thresholds are met.
- It is allowed to generate bounded evidence and exposure changes. It is not allowed to rewrite answer-bank weights, pair-quantum outputs, polarity axes, archetype labels, or workbook-owned score law.
- ExsAnonymous, Last Light, Stillness / Selah, and related Sanctuary flows are strong grief-producing sources, but they remain governed signal producers. A heavy session does not automatically become a permanent identity fact.

2. Architectural position inside Genesis OS

The corrected blueprint already treats companion systems as controlled signal producers with influence caps. This contract makes E_GRIEF one of those governed companions and defines exactly how it plugs into Sanctuary, AI orchestration, Human Emotion Genome, readiness, forecast, moderation-safe display, and telemetry.

Layer	Role of E_GRIEF	Allowed outputs	Not allowed to do
Sanctuary runtime	Adaptive regulation and grief-aware pacing	entry posture, auto-Selah, softer prompts, session gating	force a user into therapy-like flows or block all expression
AI orchestration	Tone, cadence, and prompt guardrails	presence mode, slowed questioning, gentler reflection framing	invent diagnoses or overstate certainty
Human Emotion Genome	Fast + medium pattern evidence under policy	feature packets, grief-state observations, recovery-trajectory features	overwrite stable identity without governed jobs
Readiness / forecast context	Bounded context only after thresholds	suppression hints, confidence modifiers, trend packets	directly assign final match scores or uncapped penalties
Partner / social surfaces	Safety-focused exposure filtering	soft labels, suppressed nudges, reduced social push	expose raw grief internals to partners

3. Canonical definitions

Term	Definition
Spiritual Depletion Index (SDI)	A 0-100 bounded runtime index representing acute-to-medium grief load across mind-load, body-load, and spirit-load signals.
E_GRIEF state	The governed grief runtime state derived from SDI tier, trend direction, and recency.
Grief event	A logged event indicating a grief-relevant signal or

	session outcome with provenance and confidence.
Grief episode	A grouped window of grief events representing one active grief period for a user.
Influence cap	A hard boundary limiting what grief can change downstream, how strongly, and for how long.
Runtime exposure rule	A policy rule that determines which product surfaces, prompts, AI modes, or notifications may be shown at each SDI tier.
Recovery trajectory	A governed trend description showing whether grief load is easing, stable, returning, or escalating across time.

4. Canonical signal families and source registry

E_GRIEF only uses explicit source families. Every incoming grief signal must carry provenance, event timestamp, source family, source object id, and rules version.

Source family	Examples	Evidence class	Can create grief event alone?	Typical use
Sanctuary / Stillness	Selah duration, repeat stillness, frozen visits, open-then-leave	direct	yes, with thresholds	acute state detection
ExsAnonymous	completed session, ex-story capture, residue pass, closure loop	direct	yes	high-value grief session input
Last Light	ritual completion, memorial inputs, release ritual	direct	yes	high-value grief session input
Whisper Window	loss-oriented drafts, goodbye framing, unsent grief expression	direct + contextual	sometimes	language enrichment
Messaging / social avoidance	sudden avoidance after grief-heavy session, suppressed engagement	contextual	no	supporting evidence only
Language extractor	loss words, finality, regret loops, self-blame, spiritual wrestling	direct	yes, if threshold met	fine-grained signal detection
Usage pattern telemetry	open-leave loops, reduced interaction after intense session	contextual	no	body-load and freeze proxies
Admin / safety review	manual review note or protected appeal outcome	restricted	no, except override path	audit and correction only

5. SDI scoring logic and influence philosophy

The upgraded model already provides the seed math. This contract keeps the seed idea but productionizes it with bounded evidence classes, anti-runaway rules, and packet versioning.

Signal bucket	Production examples	Nominal score band	Production note
Session score	ExsAnonymous complete, Last Light complete, both	10-35	High-value direct signals;

	within 7 days		versioned event weights
Language score	loss/finality, regret loops, self-blame, spiritual wrestling	0-25	Requires extractor thresholds and exclusion checks
Stillness score	Selah above baseline, repeated Selah, post-session stillness	0-20	Compared against user baseline when available
Usage pattern score	frozen visits, open-leave loops, temporary avoidance after intense sessions	0-20	Contextual support only; cannot dominate alone
Recovery offset	contrary evidence, stable regulation, non-grief usage normalization	-20 to 0	Used for decay and recovery trajectory

- SDI_raw = bounded sum of source bucket contributions after de-duplication, exclusions, and repeat-window control.
- SDI_current = clamp(0,100, smoothed SDI_raw plus governed decay / recovery offsets).
- No single weak contextual source may drive a user into the highest grief tier alone.
- High-tier transitions require either one strong direct ritual source plus support, or multi-source confirmation within the active time window.

6. SDI tiers, semantics, and mandatory runtime exposure rules

The core upgrade introduced four practical tiers. This contract locks their meaning and exactly what each tier may change at runtime.

Tier	SDI range	System label	What it means	Mandatory runtime posture	Prohibited behavior
T0	0-20	Stable	No active grief burden beyond normal variance.	Normal Sanctuary posture. Standard copy. No grief-specific suppression.	Do not force grief prompts or silent-mode behavior.
T1	21-40	Tender	Early or mild grief load is present; user may need gentler pacing.	Gentle Selah offers, softer AI phrasing, low-pressure reflections, reduced urgency in copy.	Do not overexpose grief labels to partners or public surfaces.
T2	41-70	Active Grief	Meaningful grief burden is affecting runtime behavior and care posture.	Auto-Selah after heavy sessions, suppress social pushes, Whisper Window favors expression over decisions, slower AI cadence.	Do not trigger high-energy nudges, date-pressure prompts, or performance gamification.
T3	71-100	Deep Grief	User is in an acute or spirit-heavy grief state requiring maximum softness and reduced pressure.	Sanctuary defaults to stillness-first, notifications reduced, AI enters ultra-soft presence, ExsAnonymous and Last Light	Do not push matching, partner-pressure language, streak mechanics, or aggressive engagement loops.

				become safe holding spaces.	

7. Runtime exposure matrix by module and surface

Runtime exposure rules define what the app is allowed to show, auto-trigger, suppress, or restyle. This is the heart of production behavior. It keeps grief adaptation useful without letting the engine turn into a black-box puppet master.

Surface / module	T0 Stable	T1 Tender	T2 Active Grief	T3 Deep Grief
Sanctuary home	standard entry	soft entry banner optional	stillness-forward card on top	stillness-first landing by default
Stillness / Selah	optional	gentle nudge if recent heavy signal	auto-offer after heavy sessions	default recommended start path
Whisper Window	standard mode	grief-safe language suggestions	expression-first mode; no decision pressure	ultra-soft expression mode; prompt count reduced
ExsAnonymous	available normally	highlighted as supportive option	safe holding mode; reduced challenge prompts	safe holding mode only; no hard confrontational prompts
Last Light	available when relevant	gently surfaced if memorial cues present	protected ritual framing	protected ritual framing + reduced surrounding noise
Mirror Matches / social prompts	normal	slightly softened outreach pressure	social push suppressed	social push suppressed
Notifications	normal cadence	reduced intensity	reduced promotional/social cadence	minimal non-essential notifications
AI coach / reflection	standard reflective tone	soft, validating cadence	slow cadence, no optimization pressure	ultra-soft presence, low-demand questions
Partner-facing surfaces	normal product rules	no raw grief tier exposure	no raw grief tier exposure	no raw grief tier exposure

8. Influence caps and downstream permissions

Influence caps are the anti-chaos law. They define exactly how far E_GRIEF may reach into other systems. The grief engine matters, but it does not get to hijack everything because one ritual hit hard.

Downstream area	Allowed influence	Cap type	Duration / TTL	Requires threshold?
Sanctuary UI posture	module ordering, copy softness, default entry path	direct runtime cap	session to 7 days depending on tier	yes
AI tone and cadence	prompt style, pacing, suggestion restraint	direct runtime cap	session to 14 days	yes
Readiness context	bounded softness / suppression hint, confidence modifier	context packet only	7-30 days	T2+ and sufficient evidence
Forecast context	trend packet or caution context only	context packet only	14-45 days	T2+ and multi-source evidence
Match / social nudges	suppression or soft throttling	behavioral exposure cap	up to 30 days	T2+ or T3
Stable identity / archetype / shadow	none except evidence note for governed jobs	hard no-overwrite cap	n/a	not allowed directly
Partner-visible output	soft explanatory band	hard privacy cap	n/a	default no raw tier

	only if explicitly allowed later			

- E_GRIEF may influence experience design strongly and score law weakly.
- It may create context packets and suppression hints, but final readiness or forecast numbers remain under their own contracts.
- If grief evidence conflicts with stable assessment signals, the runtime must favor temporary care posture rather than rewriting stable identity.
- Any request to expose raw SDI numerics outside admin tooling should be treated as a contract change, not a casual UI tweak.

9. Production packet schemas

This section converts the upgraded model into actual engineering packets. Every production write must be structured, versioned, and auditable.

Packet	Core fields	Purpose	Write target
grief_signal_event	event_id, user_id, source_family, source_object_id, signal_family, raw_strength, confidence_band, occurred_at, rules_version	Atomic grief signal record	event ledger
grief_session_summary	session_id, user_id, module_id, start_at, end_at, session_intensity, language_hits, stillness_delta, post_session_freeze_flags	Module/session rollup	Sanctuary / module store + ledger
grief_state_snapshot	user_id, sdi_current, sdi_tier, trend_dir, episode_id, computed_at, decay_at, evidence_count_window, exposure_profile_id	Current runtime state	state store
grief_context_packet	user_id, tier, trend, confidence_band, allowed_runtime_actions, suppressed_actions, ai_mode, ttl, source_refs	Downstream runtime packet	AI orchestration + app runtime
grief_feature_packet	user_id, lane, feature_key, value_norm, confidence_band, freshness_window, source_refs, validator_result	Genome / feature store output	feature store
grief_admin_audit	audit_id, subject_user_id, event_refs, reason_code, reviewer_action, before_state, after_state	Manual review / correction	admin audit store

10. Event catalog and signal families

The event catalog prevents grief detection from becoming vague or overbroad. Each event name below should exist in code tables with versioning, enabled flag, and default threshold metadata.

Event code	Description	Class	Emission timing	Notes
GRF_EXS_COMPLETE	ExsAnonymous	direct	session end	Strong ritual grief

	session completed			source
GRF_LASTLIGHT_COMPLETE	Last Light ritual completed	direct	session end	Strong memorial / release source
GRF_LOSS_LANGUAGE_HIT	Loss or finality language threshold crossed	direct	message / prompt response parse	Needs extractor threshold and exclusions
GRF_REGRET_LOOP_HIT	Regret / self-blame threshold crossed	direct	parse + repeat window	Mind-load signal
GRF_SPIRIT_WRESTLE_HIT	Spiritual wrestling threshold crossed	direct	parse + repeat window	Spirit-load signal
GRF_SELAH_ABOVE_BASELINE	Stillness duration exceeds baseline band	direct	session end	Body + spirit load support
GRF_REPEAT_SELAH_24H	Repeated stillness / Selah within 24h window	direct	windowed job	Acute burden support
GRF_FROZEN_VISIT	Open / stall / no-action pattern after heavy emotional surface	contextual	usage telemetry job	Support only
GRF_OPEN_LEAVE_LOOP	Open then leave loop repeated	contextual	usage telemetry job	Support only
GRF_SOCIAL_AVOID_AFTER_HEAVY	Avoids social or matching surfaces after heavy session	contextual	windowed job	Support only
GRF_RECOVERY_NORMALIZATION	Contrary recovery signal reduces active burden	recovery	windowed job	Decay / recovery offset
GRF_ADMIN_CORRECTION	Manual correction or protected override	restricted	admin action	Audit-only except approved correction

11. State machine and episode logic

E_GRIEF needs both an event stream and a user-level state machine. One emotional moment is not a lifetime sentence. The state machine exists to group, soften, decay, and, when appropriate, resolve grief burden.

State	Entry conditions	Primary behavior	Exit conditions
Idle / Stable	SDI in T0 and no active episode	Standard product posture	New qualifying events raise SDI
Tender monitoring	SDI enters T1	Soft pacing, gentle nudge eligibility	Decay to T0 or rise to T2
Active grief episode	SDI enters T2 with threshold evidence	Suppression of social pushes; stronger Sanctuary care posture	Decay to T1/T0 or escalate to T3
Deep grief episode	SDI enters T3 with strong evidence	Stillness-first, ultra-soft presence, maximum pressure reduction	Decay to T2/T1 with contrary evidence and time
Recovery watch	Post-T2/T3 downward trend	Continue softer posture until stable enough	Return to T0 or re-open episode on new evidence
Resolved archive	Episode closed after decay and contrary evidence	Historical packet only	New episode opens on later evidence

- Episode open rule: one strong direct ritual event plus support, or repeated qualifying direct signals within the active window.

- Episode reopen rule: new direct grief events within the recovery-watch window may re-open the prior episode if continuity is likely.
- Episode merge rule: adjacent events within configured continuity windows may merge into one episode to avoid fragment spam.
- Episode close rule: sufficient decay, contrary evidence, and absence of new high-weight events.

12. Runtime exposure rules by AI mode

AI behavior should feel meaningfully safer, not dramatically different in personality every hour. This contract therefore defines exposure by mode rather than by random prompt creativity.

Tier / mode	Cadence	Prompt count	Question style	What AI should avoid
T0 / Standard	normal	normal	balanced reflective prompts	nothing beyond general policy
T1 / Soft presence	slower	slightly reduced	gentle naming and grounding prompts	hard challenges, optimization pressure, bright hype language
T2 / Expression-first	slow	reduced	containment, expression, simple reflection	decision pressure, dating urgency, performance framing
T3 / Ultra-soft presence	very slow	minimal	holding, stillness, low-demand companionship	debate, challenge loops, streak pressure, prescriptive certainty

13. Visibility and privacy rules

Grief is not a badge to pin on the user. Production behavior must distinguish between internal state, user-facing support, and partner-safe visibility.

Audience	May see raw SDI?	May see tier label?	Allowed phrasing	Default
User self surface	No raw numeric by default	Soft label if designed later	'You may need a softer pace right now.'	soft explanatory copy only
Partner / match	No	No raw tier	At most system-safe phrases like 'this app is slowing social pressure right now.'	suppressed
Admin / audit	Yes	Yes	full audit language	allowed with permissions
AI internal orchestration	Yes via packet	Yes	structured runtime fields, not melodramatic prose	allowed under service auth
Model training / analytics	featureized, de-identified where required	aggregated only	technical analytics labels	allowed under data policy

14. Rollout boundaries and launch sequencing

The upgraded module is usable now, but production rollout must be staged. This keeps grief adaptation from touching too many surfaces before the telemetry and audit trail are proven.

Phase	Allowed scope	What ships	What stays off
Phase 1 - Internal validation	staff / staging only	event catalog, state	partner-facing anything,

		snapshots, AI tone modes, Sanctuary logging	readiness/forecast downstream writes
Phase 2 - Private beta	limited user cohort	Sanctuary runtime adaptation, Selah triggers, ExsAnonymous and Last Light grief packets	raw numeric display, forecast packets, broad notification suppression
Phase 3 - Controlled production	opt-in or governed rollout	full exposure matrix for Sanctuary + AI + social throttling, genome features	partner-visible grief labels, unrestricted admin overrides
Phase 4 - Mature production	after audit confidence	bounded readiness / forecast context packets and richer recovery trajectory tools	any uncapped score mutation or public raw SDI display

15. Data objects and code tables that must exist

This module should not live as prose only. The following code tables and stores are required so the system can actually behave the same way across backend, AI, analytics, and app layers.

- grief_event_code registry
- grief_signal_family registry
- grief_tier registry with exposure_profile_id linkage
- grief_exposure_profile table
- grief_decay_policy table
- grief_episode store
- grief_state_snapshot store
- grief_context_packet store / cache
- grief_feature_key registry for Genome writes
- grief_copy_profile registry for user-safe language
- grief_audit_reason registry for manual interventions

16. Example feature keys for Human Emotion Genome

Feature key	Lane	Meaning	Write posture
grief.sdi.current	fast	Current bounded grief load	snapshot only
grief.sdi.tier	fast	Current grief tier	snapshot only
grief.episode.active	fast	Whether a grief episode is active	snapshot only
grief.episode.count_90d	medium pattern	How many episodes in 90 days	bounded aggregate
grief.ritual.lastlight.count_30d	medium pattern	Last Light usage cadence	aggregate only
grief.ritual.exs.count_30d	medium pattern	ExsAnonymous cadence	aggregate only
grief.language.loss.band	fast/medium	Loss-language intensity band	bounded feature
grief.recovery.trajectory	medium pattern	easing / stable / returning / escalating	derived feature
grief.social.throttle.active	fast	Whether social push suppression is active	runtime flag
grief.context.confidence	fast	Confidence band for grief packet	snapshot only

17. API contract summary

The app and services need a simple API surface, even if internals are event-driven.

Operation	Input	Output	Used by
POST /grief/events	signal or session event payload	accepted event_id + processing status	Sanctuary, AI, telemetry jobs
GET /grief/state/{user_id}	user id + auth context	current state snapshot + exposure profile	app runtime, AI orchestration
POST /grief/recompute	user id or episode id	new snapshot + audit metadata	jobs / admin tools
POST /grief/admin-correction	audit payload	before/after state + audit id	restricted admin
GET /grief/exposure-profile	tier or profile id	runtime exposure rules	client runtime / orchestration

18. QA, audit, and false-positive control

Grief detection should be compassionate and deterministic. It cannot be sloppy. The QA suite should include deliberate false-positive challenges so the system does not call every difficult feeling 'grief.'

- Sadness without loss language should not automatically raise T2/T3.
- Single open-leave behavior without direct ritual or language support should remain contextual only.
- Repeated grief rituals should increase state, but de-duplication must prevent one session from multiplying across services.
- Recovery signals must actually lower burden over time when no new grief events appear.
- Partner-safe displays must never leak raw SDI or internal grief notes.
- Admin corrections must leave audit trails and must not silently erase historical events.
- AI mode shifts must match tier transitions deterministically and revert when decay rules are met.

19. Implementation checklist

- Publish code tables for tiers, event codes, exposure profiles, decay policies, and feature keys.
- Implement event ingest and de-duplication pipeline.
- Implement state snapshot job with decay and recovery offsets.
- Connect Sanctuary modules, especially ExsAnonymous and Last Light, to emit canonical grief packets.
- Wire AI orchestration to consume grief_context_packet rather than raw event spam.
- Ship rollout gates by phase; do not light up every downstream integration at once.
- Add dashboarding for tier counts, decay behavior, false positives, and suppressed social nudges.

20. Closing system law

E_GRIEF is one of the most emotionally sensitive companion systems in Genesis OS. That means it must be precise, soft, auditable, and humble. It should help the product know when to slow down, hold the user more carefully, and reduce pressure - not pretend it has become the king of the entire engine. Built correctly, E_GRIEF gives Sanctuary, ExsAnonymous, Last Light, AI, and readiness context a shared grief-aware runtime language. Built poorly, it becomes mood theater with a database. This contract exists so the dev team builds the first version, not the second one.